



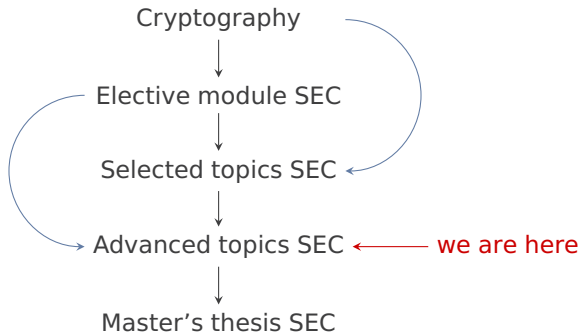
Security, Privacy, and Forensics

SE 703364: Secure and Distributed Computing

Rainer Böhme

4 March 2026

Study in Order



Requirements for Passing this Course

- Regular attendance
- Contributions to discussions
- Working on a topic

Topic area in this term: Are AI vulnerabilities special ?

Each topic is presented in three steps:

1. “Context presentation” of related work;

This presentation ends with a problem statement and a proposal for a small experiment of your own.

2. “Result presentation” of the (apparent) solution to the problem;

This presentation is supported with selected pieces of (own) evidence.

3. Seminar thesis documenting the process with your own focus.

Syllabus

Summer Term 2026

04.03.26	1. Kick-off	
11.03.26	No meeting (Q & A on topics and final assignment)	
18.03.26	2. Practitioners talk (tbc)	
25.03.26	3. CHI conference talk rehearsal	Nora Hofer
22.04.26	4. Status presentations (on MSc thesis projects)	Thomas Filips, (+1)
29.04.26	5. Context presentations	
06.05.26	6. Thesis defense (tbc)	
20.05.26	Reserved	
27.05.26	7. Status presentations (on MSc thesis projects)	Max Ninow, (+1)
03.06.26	Reserved	
10.06.26	8. IH conference talk rehearsal	
24.06.26	9. Result presentations	
17.07.26	Deadline for seminar thesis submission	

Last update: March 4, 2026

Criteria for Presentations

Content

Focus on central theme

Do not lose your audience

Use running examples

Avoid unnecessary detail

Prepare for questions

Practice, practice, practice !

Form

Simplicity rules

Prefer smart (own) visualization over bullet points

Equations where necessary

Have backup slides

Test contrast, color, (laptop)

Time limit: 20 min (+ Q & A)

No experiments

Criteria for Term Papers

Content	Form
Creativity of the approach	Visual appearance
Correctness	Correct citation and references
Clarity and structure	Language and spelling
Soundness of arguments	Tables and figures
Technical sophistication	Consistency of terminology and notation
Interpretation of results	Prefer active voice
No clone of the original paper	e. g. $\text{\LaTeX}$, Bib $\text{\TeX}$, TikZ

Writing a Perfect Term Paper

"I would have written a shorter letter, but I did not have the time."

Blaise Pascal

Target group: your fellow students, after the cryptography & security lectures

- Connect to lecture, but do not repeat too much

Page limit: 12

(single column, excl. references and supplemental tables/figures/code)

- Summarize your approach 2 pages
- Recall necessary background using one selected figure and essential formal notation 3 pages
- Describe method and procedure 3 pages
- Present result 2 pages
- Discuss result and conclude 2 pages

Grading

The final mark is the weighted average followed by unbiased rounding:

	%
Active participation in discussions	10
Context presentation	30
Result presentation	20
Term paper	40

Failure in one sub-grade leads to failure of the course.

Topic 1: Vibe Coding and Software Vulnerabilities

LLMs have become tools for automating programming tasks. Whether or not they introduce more or less security vulnerabilities has been extensively researched, but the results depend on advances in the models, fine-tuning, tasks, and prompts – as well as the programming language.

Present what is known (and still valid), and where research gaps remain. In particular, do the results apply to Rust, a programming language designed to avoid certain weaknesses, which may nevertheless be under-represented in the training data?

Starting literature

- Sandoval, G., Pearce, H., Nys, T., Karri, R., Garg, S., and Dolan-Gavitt, B. (2023). Lost at C: A user study on the security implications of large language model code assistants. In *USENIX Security Symposium*, pages 2205–2222, Anaheim, CA.
- Basic, E. and Giaretta, A. (2025). From vulnerabilities to remediation: A systematic literature review of LLMs in code security. Mimeo. <https://arxiv.org/abs/2412.15004>.

Communication Channel

We use the matrix room **se-security-privacy-forensics-2026** for announcements, updates, and other information exchange related to the seminar. Feel free to contribute !

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